Promote -
can you write a python script that takes input from the user and produces a hash. The hash values that I want are md-5, sha-256 and sha-512

Understand

Your Python hash program uses the `hashlib` library.

In your screenshot and uploaded Python file, you are using `hashlib` to generate three types of hashes.

Hash algorithm	Python code
MD5	`hashlib.md5(data).hexdigest()`
SHA-256	`hashlib.sha256(data).hexdigest()`
SHA-512	`hashlib.sha512(data).hexdigest()`

What does `hashlib` do?

`hashlib` is a built-in Python library that calculates hash values from text or files. You do not need to install it separately.

Your program starts with:

import hashlib

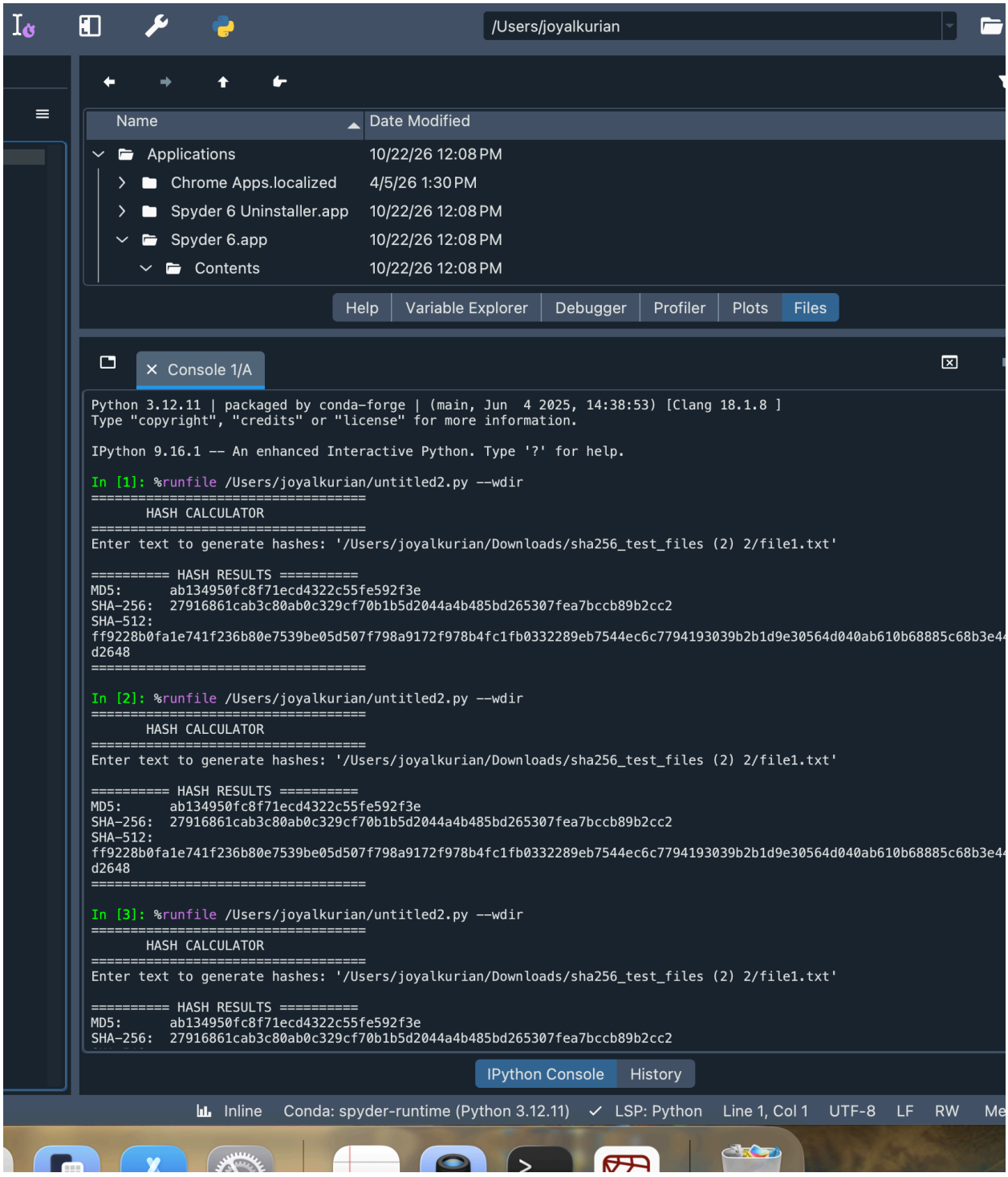
Here is how the main parts work:

1. `import hashlib` — loads Python's hashing library.

2. `user_input.encode("utf-8")` — converts the input text into bytes.
3. `hashlib.sha256(data)` — calculates the SHA-256 hash.
4. `.hexdigest()` — converts the hash into a readable hexadecimal string.

```
# Generate hash values
md5_hash = hashlib.md5(data).hexdigest()
sha256_hash = hashlib.sha256(data).hexdigest()
sha512_hash = hashlib.sha512(data).hexdigest()
```

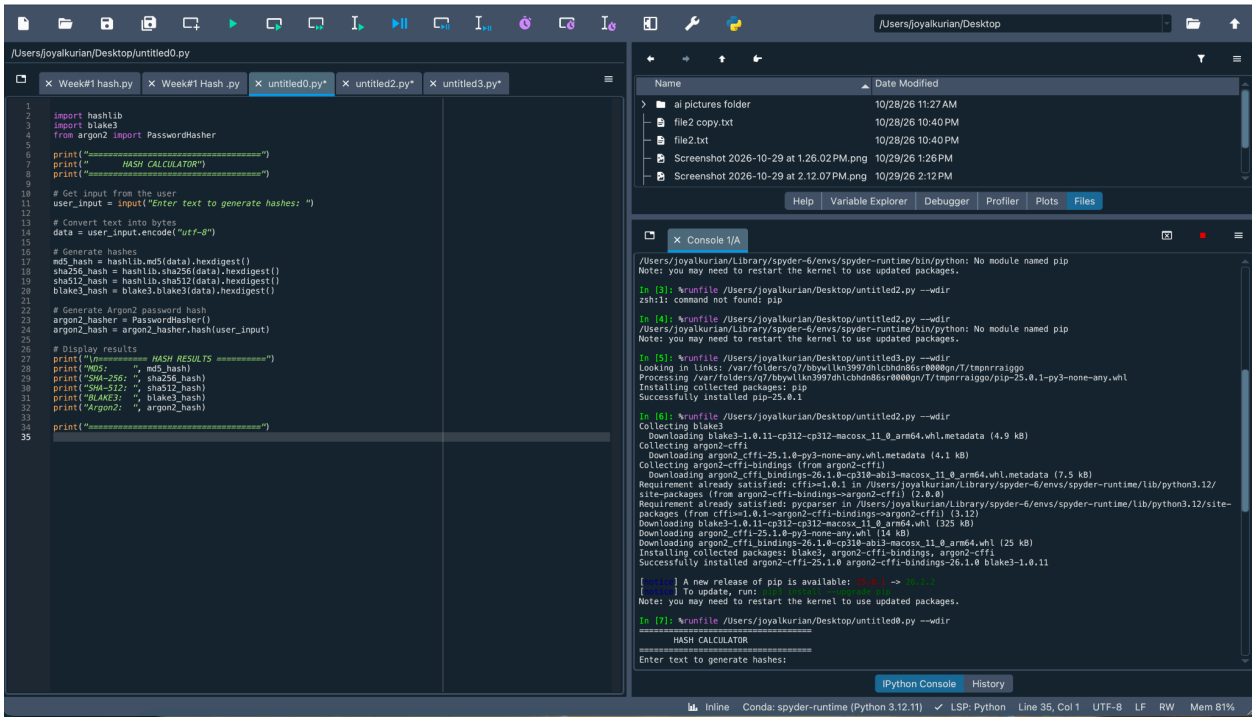
Test-



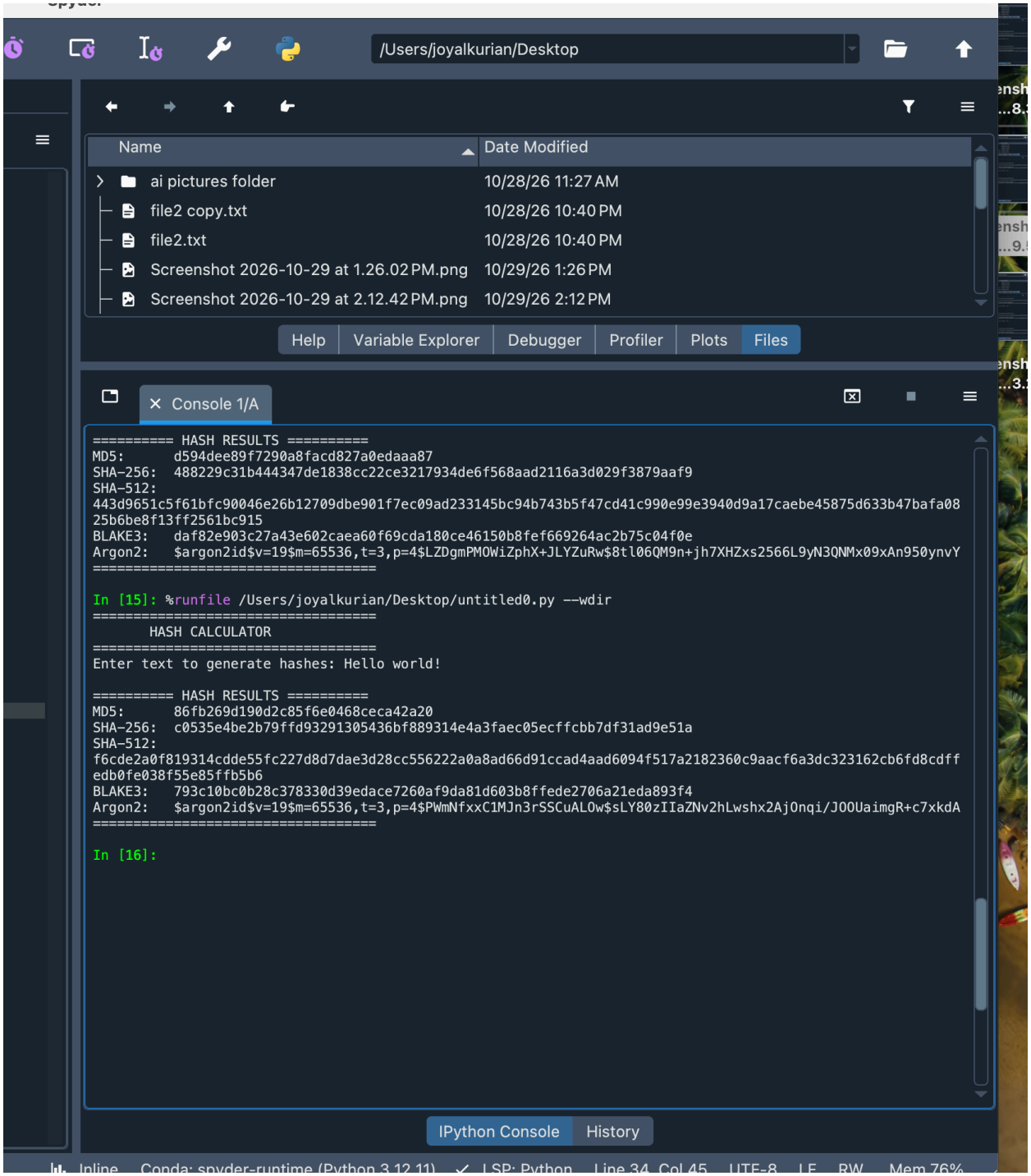
Verify-
I have checked the results about known hashes.

Security funcation-
Hashing for integrity

Critical Thinking model

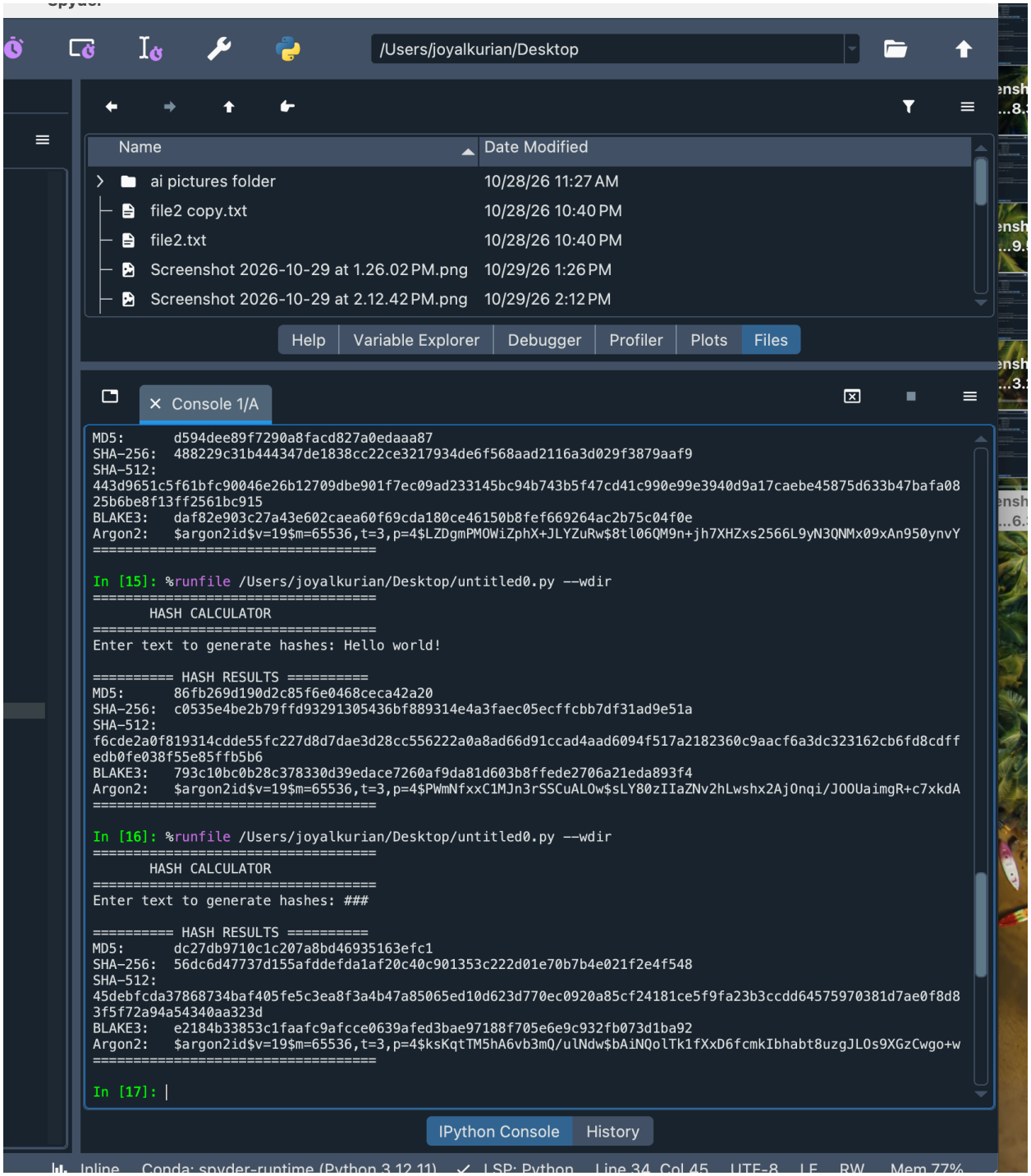


Hello world!
Hello world! (BLAKE 3) -
793c10bc0b28c378330d39edace7260af9da81d603b8ffede2706a21eda893f4
Hello world! (Argon2)- $argon2id$**v=19**$m = 19456, t = 2, p = 1$
nuwOFQPomTqghvA5JunQ4A$qTM2AZGd4ixCMSF9MY5jtrOjLa1MmyfAO49Bwwvijhs



2. ###.

(BLAKE 3)- e2184b33853c1faafc9afcce0639afed3bae97188f705e6e9c932fb073d1ba92
(Argon2)- $argon2id$**v=19**$m = 19456, t = 2, p = 1$
oDmHXUcl8Slk4jcHaBtq5g$gudlRAUl2z9hmzNZLQHS/qT30p4zcKKKnH5yw7icgYw



Part 2

Test cases

1. **Hello world!**

MD5

128 bits

86fb269d190d2c85f6e0468ceca42a20Copy

SHA-1

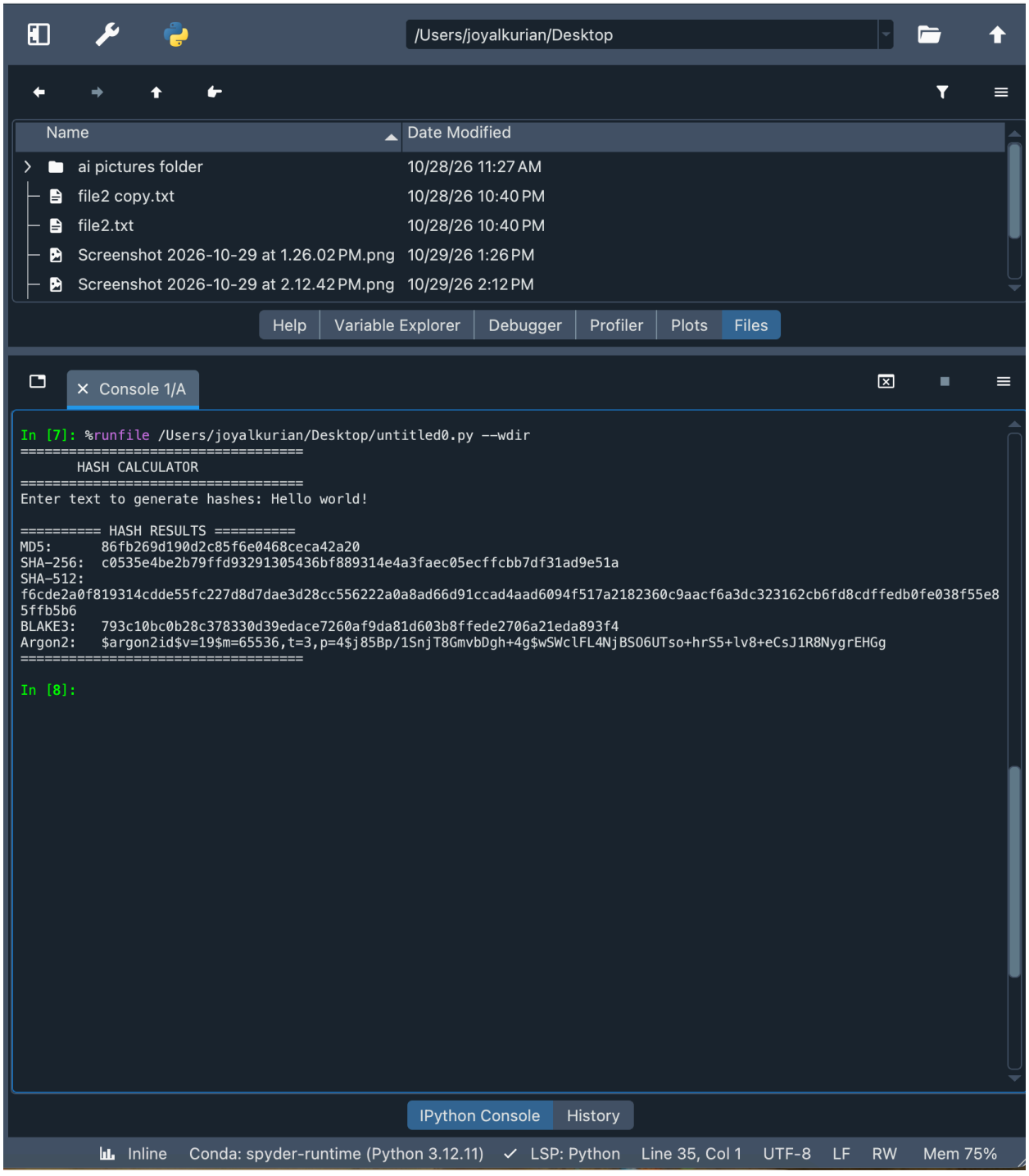
160 bits

d3486ae9136e7856bc42212385ea797094475802Copy

SHA-256

256 bits

c0535e4be2b79ffd93291305436bf889314e4a3faec05ecffcbb7df31ad9e51a



2. **23462**

MD5

128 bits

777669af68dbccabc30c3b6bcaa81825Copy

SHA-1

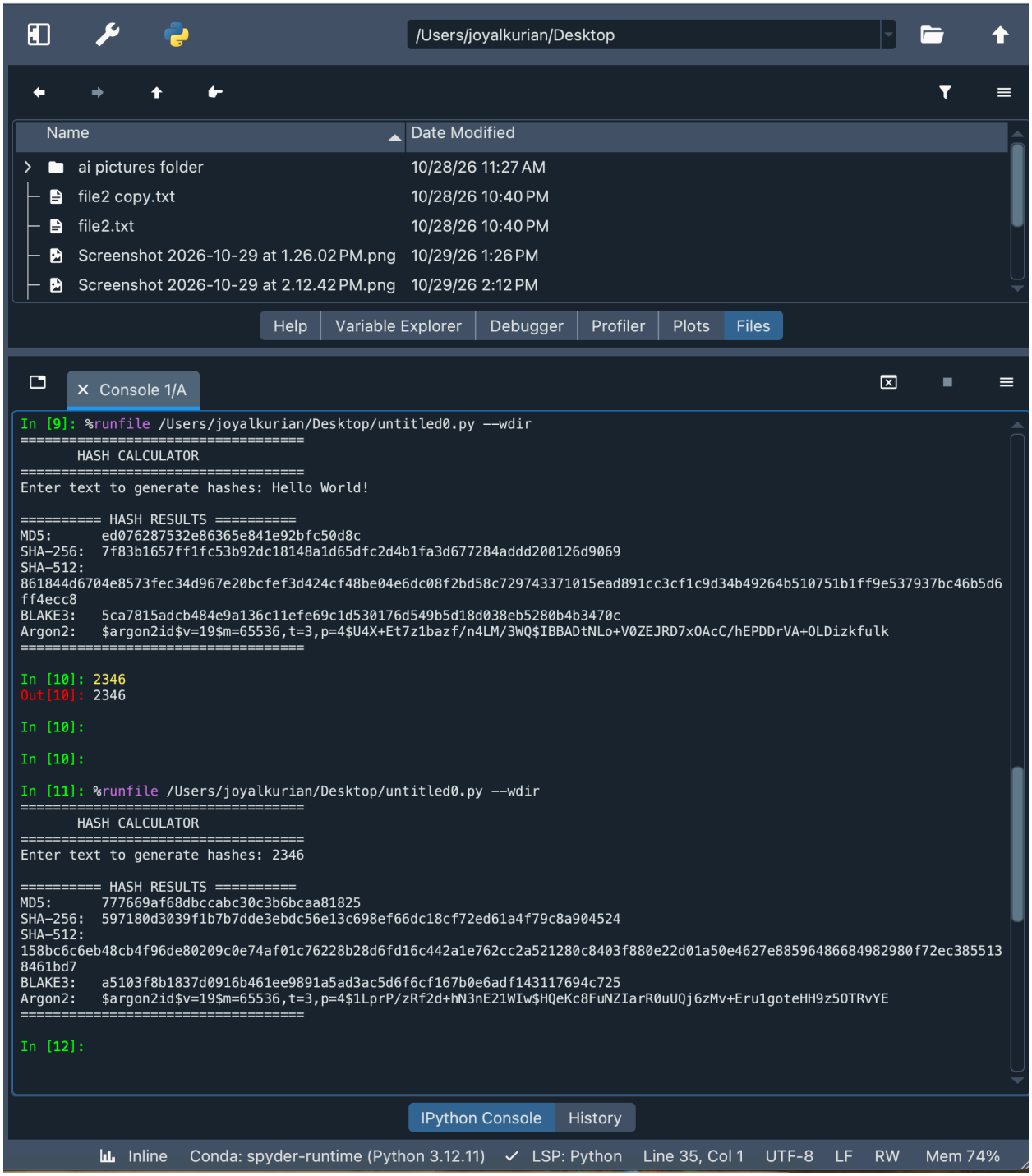
160 bits

a6e30b2e057ef7205fde876d0be2f643aa74c0cbCopy

SHA-256

256 bits

597180d3039f1b7b7dde3ebdc56e13c698ef66dc18cf72ed61a4f79c8a904524



3. ###

MD5

128 bits

dc27db9710c1c207a8bd46935163efc1Copy

SHA-1

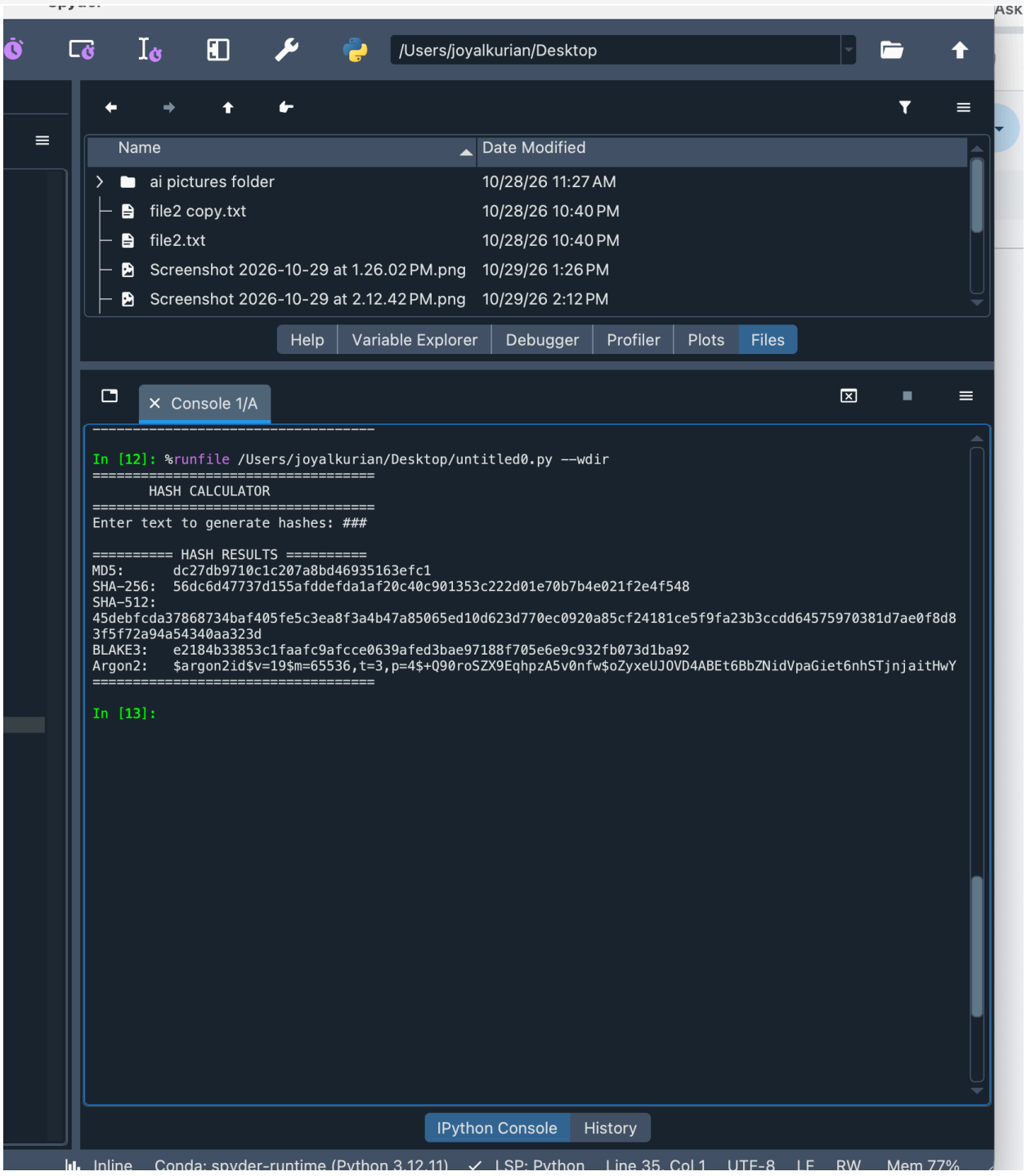
160 bits

5c5b8251bdb4b62eff8c746f4021483bfae24eb7Copy

SHA-256

256 bits

56dc6d47737d155afddefda1af20c40c901353c222d01e70b7b4e021f2e4f548



4. Holy Bible

MD5

128 bits

bd099d290044b7adf64fba0a155b9e64Copy

SHA-1

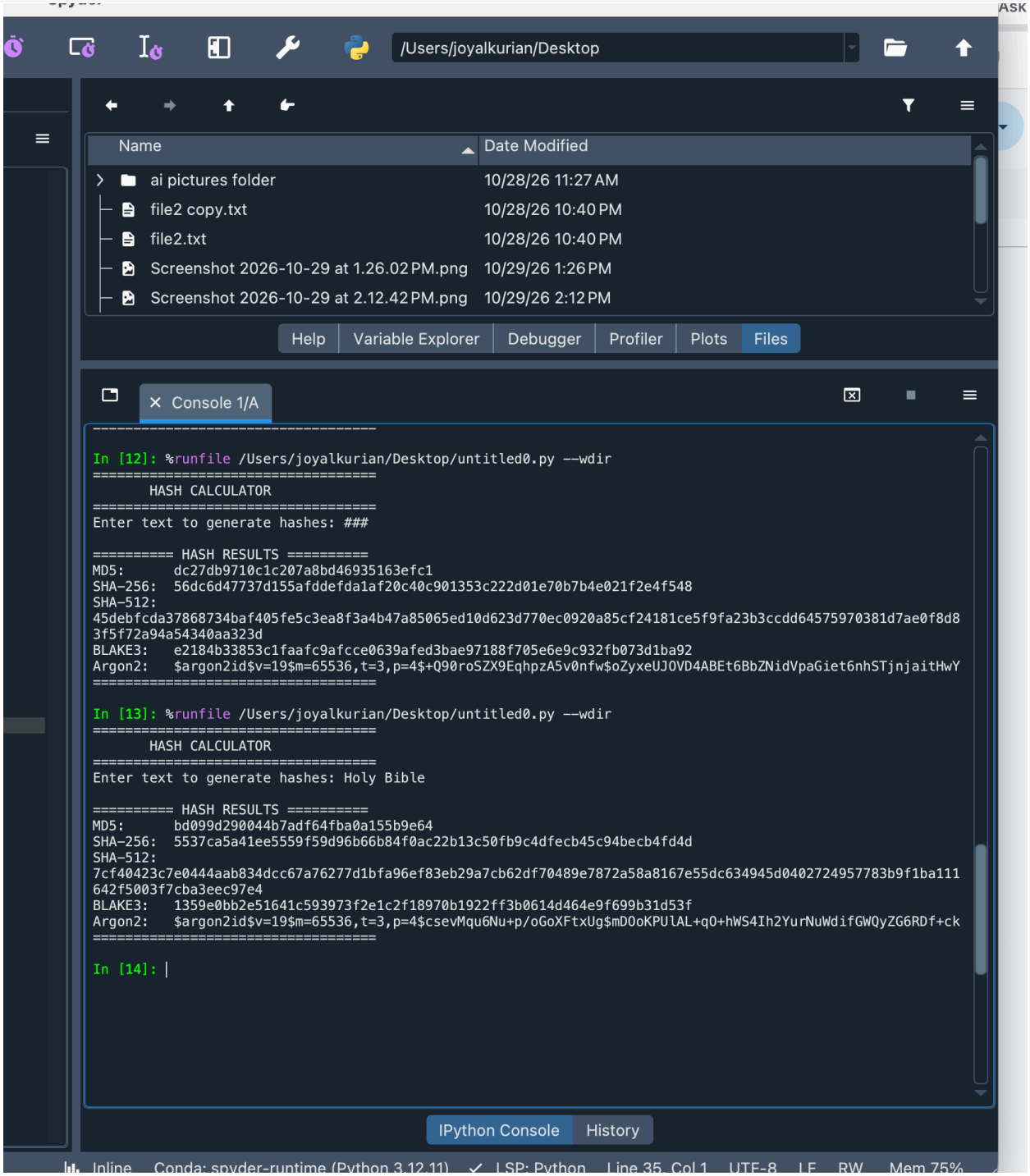
160 bits

43f028d56c114a4cd0f59bf074405821449716b4Copy

SHA-256

256 bits

5537ca5a41ee5559f59d96b66b84f0ac22b13c50fb9c4dfecb45c94becb4fd4d



5. **9497580**

MD5

128 bits

d594dee89f7290a8facd827a0edaaa87Copy

SHA-1

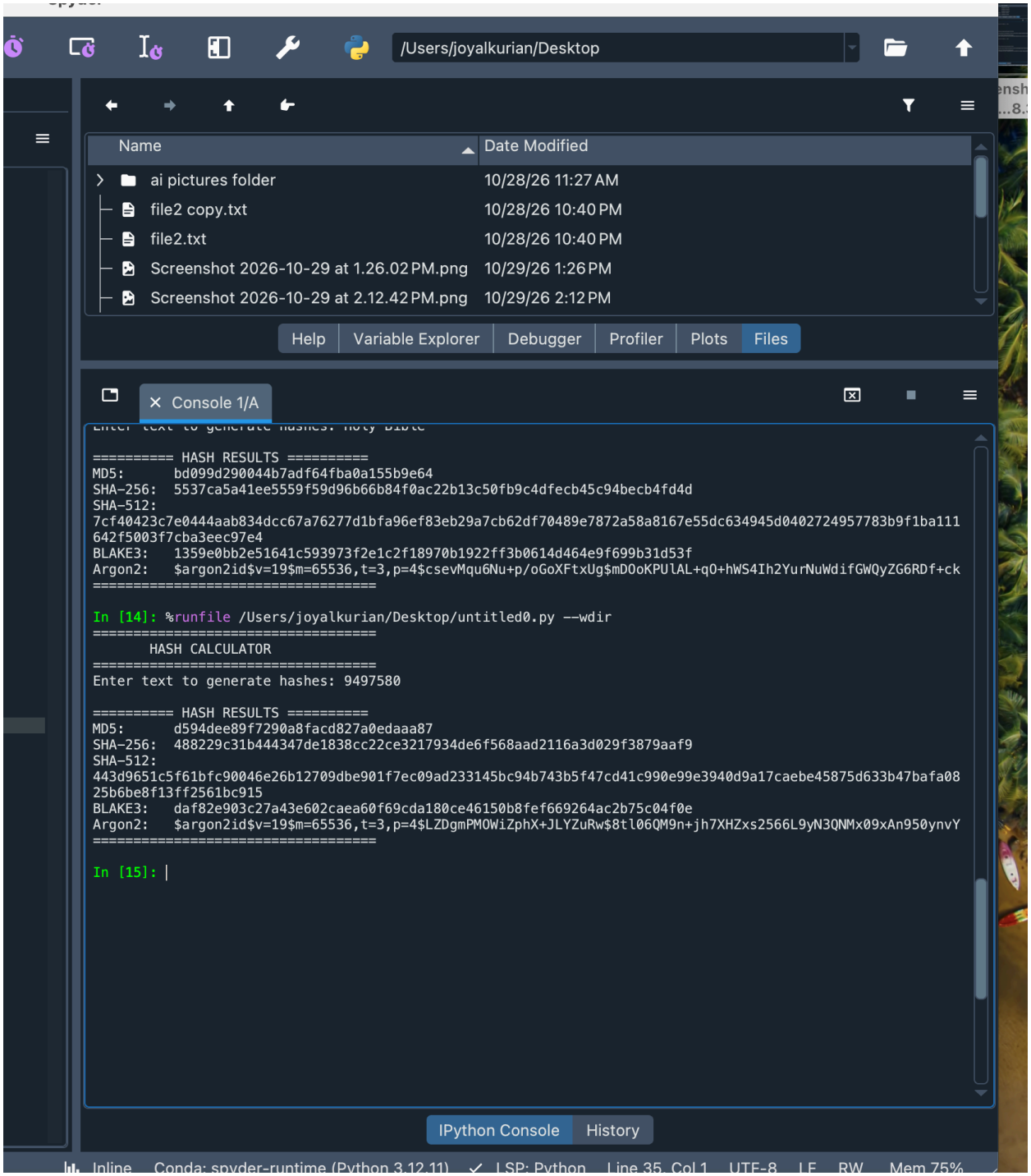
160 bits

03c72ab5d2caa287a453981482472bc09852abc3Copy

SHA-256

256 bits

488229c31b444347de1838cc22ce3217934de6f568aad2116a3d029f3879aaf9


/Users/joyalkurian/Desktop
Name
Date Modified
ai pictures folder
10/28/26 11:27 AM
file2 copy.txt
10/28/26 10:40 PM
file2.txt
10/28/26 10:40 PM
Screenshot 2026-10-29 at 1.26.02 PM.png
10/29/26 1:26 PM
Screenshot 2026-10-29 at 2.12.42 PM.png
10/29/26 2:12 PM
Help
Variable Explorer
Debugger
Profiler
Plots
Files
Console 1/A
========== HASH RESULTS ==========
MD5: bd099d290044b7adf64fba0a155b9e64
SHA-256: 5537ca5a41ee5559f59d96b66b84f0ac22b13c50fb9c4dfecb45c94becb4fd4d
SHA-512:
7cf40423c7e0444aab834dcc67a76277d1bfa96ef83eb29a7cb62df70489e7872a58a8167e55dc634945d0402724957783b9f1ba111642f5003f7cba3eec97e4
BLAKE3: 1359e0bb2e51641c593973f2e1c2f18970b1922ff3b0614d464e9f699b31d53f
Argon2: $argon2id$v=19$m=65536,t=3,p=4$csevMqu6Nu+p/oGoXFtxUg$mDOoKPUlAL+qO+hWS4Ih2YurNuWdifGWQyZG6RDf+ck
==================================
In [14]: %runfile /Users/joyalkurian/Desktop/untitled0.py --wdir
==================================
HASH CALCULATOR
==================================
Enter text to generate hashes: 9497580
========== HASH RESULTS ==========
MD5: d594dee89f7290a8facd827a0edaaa87
SHA-256: 488229c31b444347de1838cc22ce3217934de6f568aad2116a3d029f3879aaf9
SHA-512:
443d9651c5f61bfc90046e26b12709dbe901f7ec09ad233145bc94b743b5f47cd41c990e99e3940d9a17caebe45875d633b47bafa0825b6be8f13ff2561bc915
BLAKE3: daf82e903c27a43e602caea60f69cda180ce46150b8fef669264ac2b75c04f0e
Argon2: $argon2id$v=19$m=65536,t=3,p=4$LZDgmPMOWiZphX+JLYZuRw$8tl06QM9n+jh7XHZxs2566L9yN3QNMx09xAn950ynvY
==================================
In [15]:
IPython Console
History
Inline
Conda: spyder-runtime (Python 3.12.11)
LSP: Python
Line 35, Col 1
UTF-8
LF
RW
Mem 75%